

「建置臺灣空氣品質模式數據平台」計畫 教育訓練：WRF 簡介

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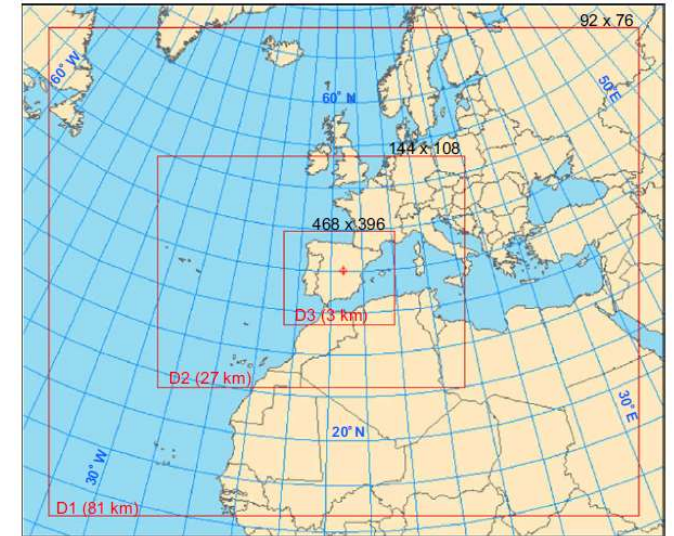
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May 2023

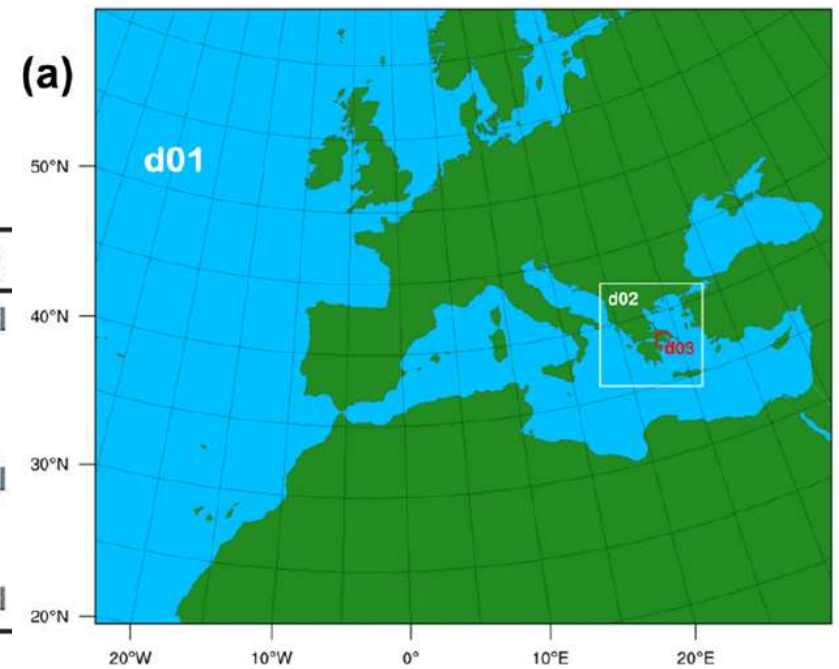
Why WRF?

1. Borge 等(2008, AtmEnv): WRFv2.2, 27-9-3km, 31層
2005 2/19-28, 6/18-27, 2-days spin-up
YSU PBL, WSM-6, Noah land-surface,
Eta GFDL longwave, MM5 shortwave radiation
time-varying SST, grid-observational nudging



2. Banks 等(2016, AtmRes): WRFv3.4.1, 12-4-1km, 38層(13層<3km, 50hPa top)
each 36-h run with first 12-h spin-up
“non-local PBL schemes give the most agreeable solutions when compared with observations”

PBL scheme	Short name	Closure type
Yonsei University	YSU	1.0 non-local
Mellor-Yamada-Janjic	MYJ	1.5 local
Quasi-normal scale elimination	QNSE	1.5 local
Mellor-Yamada-Nakanishi-Niino level 2.5	MYNN2	1.5 local
Asymmetric Convective Model	ACM2	1.0 non-local
Bougeault-Lacarrère	BouLac	1.5 local
University of Washington	UW	1.5 local
Total energy-mass flux	TEMF	1.5 non-local

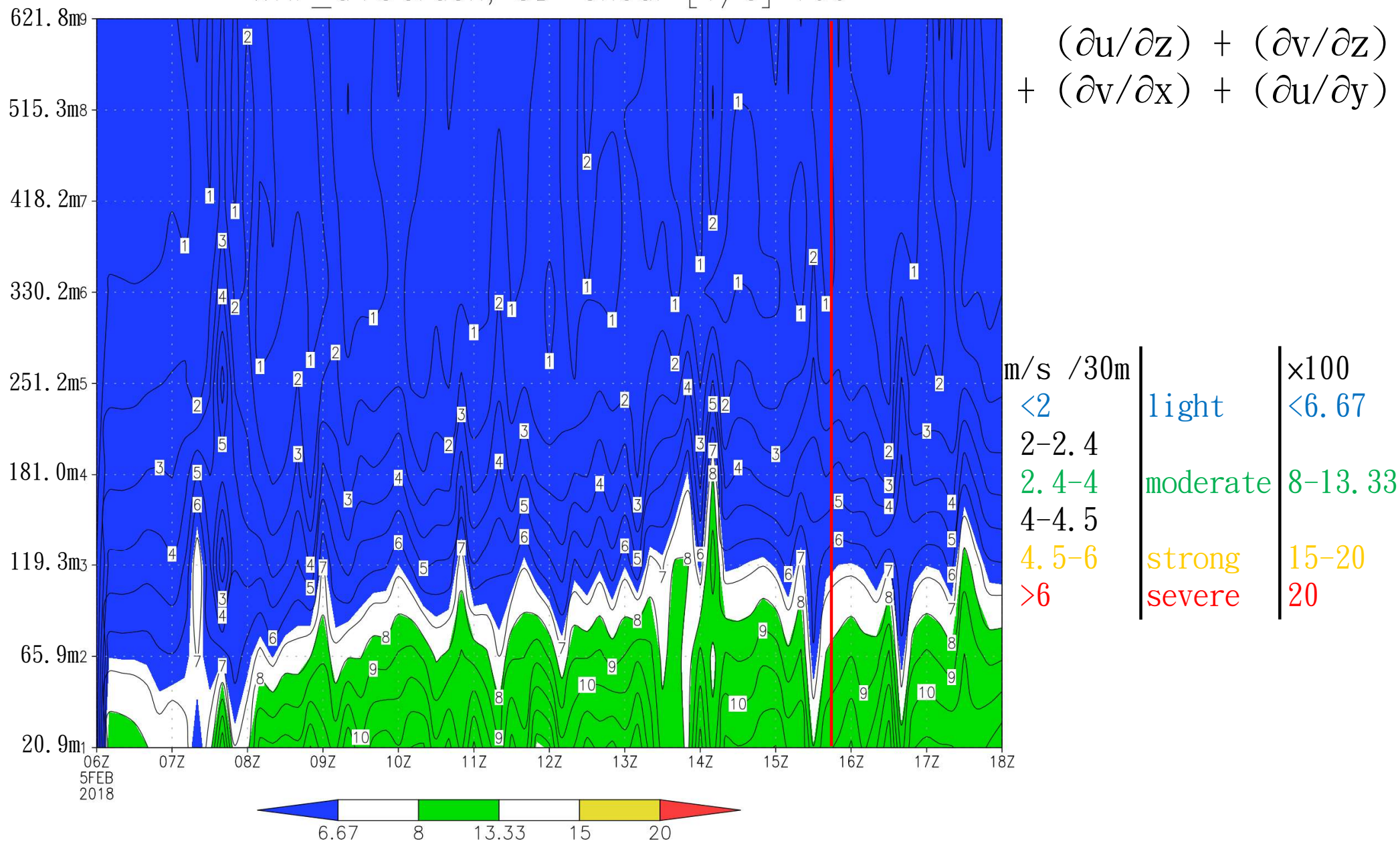


案例：02/05 2018 蘭嶼黑鷹直昇機事故 (WRFv3.9.1.1)

WRF d4: 200-m, at crash point (121.535°E, 22.026°N)

WRF grid: X:143=121.535°E, Y:143=22.026°N, AGL=0.41m

WRF_d4@crash, 3D-shear $[1/s]*100$

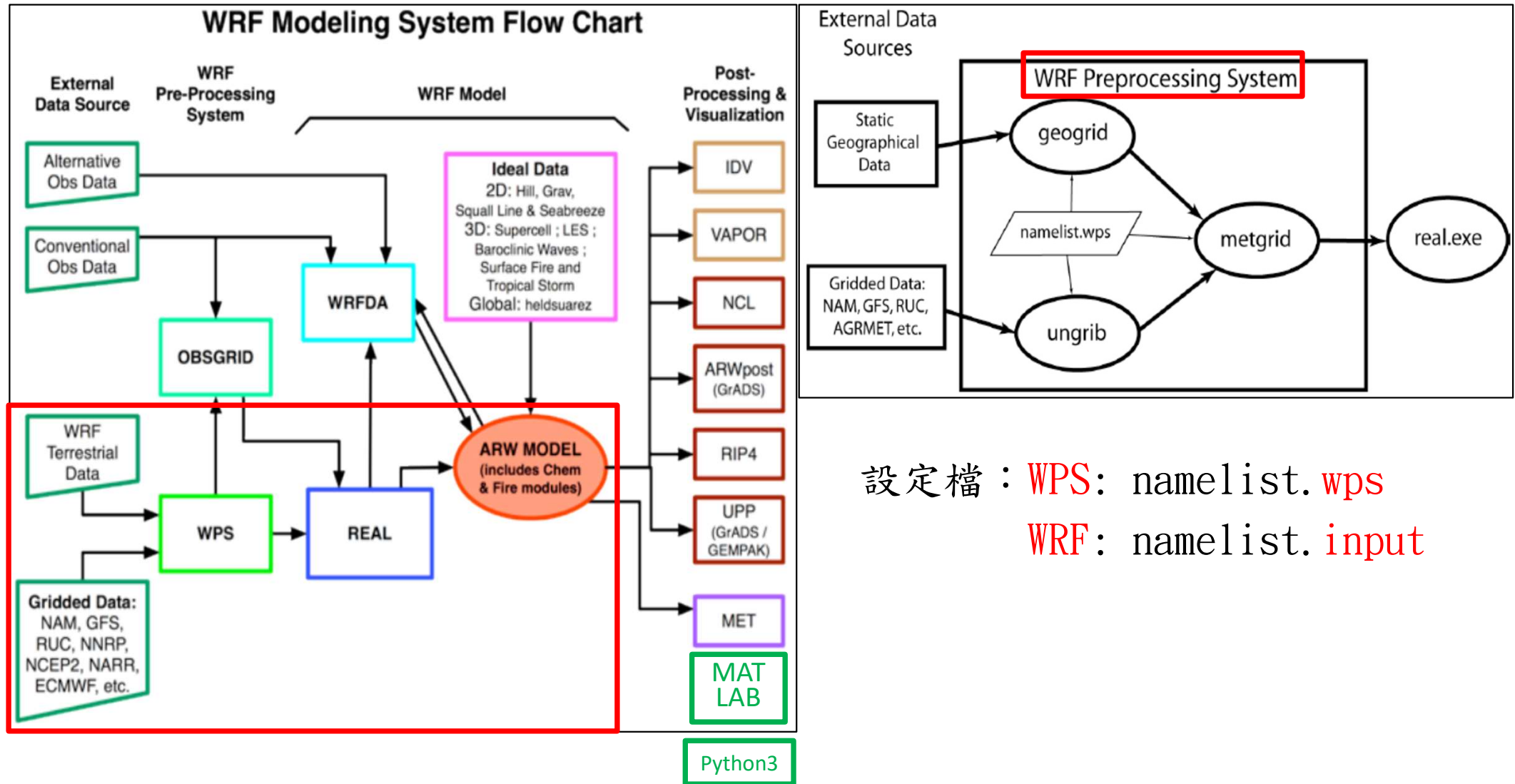


WRF 數值天氣模式

1. 分析天氣時，需要四維氣象資料而缺／無觀測資料
例：遠離機場、遠離都卜勒雷達
2. 利用中尺度天氣數值模式模擬重建氣象場
the Weather Research and Forecasting model
地形、海陸分布、輻射效應、動量方程、熱力方程、水氣效應
3. 模式預報、模擬與實際觀測之差異
數值方法、觀測誤差、可預報度……等等

Weather Research and Forecasting model

1. <https://www.mmm.ucar.edu/weather-research-and-forecasting-model>



University Corporation for Atmospheric Research manages
National Center for Atmospheric Research
on behalf of the National Science Foundation.

Weather Research and Forecasting model

2. <https://www2.mmm.ucar.edu/wrf/users/>

WRFUSERSPAGE

HomeModel SystemUser SupportDownloadDoc / PubPhysicsSupport ForumWRF ForecastLinks

**USER'S GUIDES FOR THE
ADVANCED RESEARCH WRF MODELING SYSTEM, VERSION 4**

The following is Version 4 of the [User's Guide](#) in web (html) format, and in .pdf format. If you wish to locate Versions 3.8, 3.9, and 3.9.1, please click [here](#).

Pubs & Docs Overview
Technical Description of the ARW
ARW User's Guide
Tutorial Presentations
AFWA Diagnostics Description
RIP4 Document
WRF SKEBS Document
WRF Model Code Document

Version	Web Document	PDF Document
Version 4.3	Version 4.3 UG - WEB	Version 4.3 UG - PDF * Download the tar file and unpack it. This will give a .pdf of each chapter.
Version 4.2	Version 4.2 UG - WEB	Version 4.2 UG - PDF
Version 4.1	Version 4.1 UG - WEB	No V4.1 UG PDF Version Available
Version 4.0	Version 4.0 UG - WEB	Version 4.0 UG - PDF

updated 09/29/2021 02:51:12

ARWUsersGuideV3.8.pdf ~ 39 MB

Weather Research and Forecasting model

2. <https://www2.mmm.ucar.edu/wrf/users/>

The screenshot shows the 'WRF USERS PAGE' with a navigation bar containing links: Home, Model System, User Support, Download, Doc / Pub, Physics, Support Forum, WRF Forecast, and Links. The 'Download' link is highlighted with a red box. A red dashed arrow points from this box to the '1. WRF SOURCE CODES AND GRAPHICS SOFTWARE DOWNLOADS' section on the right. The main content area is titled 'WRF MODEL USERS' PAGE' and includes a welcome message, a list of applications (Meteorological studies, Real-time NWP, Idealized simulations, Data assimilation, Earth system model coupling, Model training and educational support), information about the Mesoscale and Microscale Meteorology Laboratory (MMM), and a section for 'Related Systems and Information'. On the right side of the page, there are several informational boxes: 'WRF FORECAST' with a map, 'ANNOUNCEMENTS' with links to a tutorial and workshop, 'Latest WRF Release' (Version 4.3), 'WRF Releases - Known Problems', and 'GENERAL INFORMATION' with links to a support statement, notes, FAQs, code repository, and physics review process.

1. WRF SOURCE CODES AND GRAPHICS SOFTWARE DOWNLOADS

Before you download the WRF software, please take a minute to read the [WRF Public Domain Notice](#), and then fill-out the registration form by clicking on the link to '[New Users](#)' below. When the registration form is submitted, you will be prompted to the software download page. It also subscribes you to the [WRF news email list](#) (with your confirmation). We will be using this list to broadcast any messages regarding WRF updates and events, in addition to field-related job announcements.

If you have registered and wish to download the software again, please click on the link for '[Returning Users](#)'. You will be asked to fill-in your email address and will then be allowed to proceed.

If you have used a version prior to version 3 in the past, and wish to use version 3.0 or later, you will be asked to register again.

If you are a first-time WRF user, learn how to run the programs via the [online tutorial](#)

To DOWNLOAD, click one of the links below:

[New Users](#)

[Returning Users](#)

2. WPS V4 GEOGRAPHICAL STATIC DATA DOWNLOADS PAGE

3.

A number of datasets which can be used as input to WPS can now be downloaded directly from the [NCAR Research Data Archive \(NCAR RDA\)](#) web site:

- You must register with a user name and password to access the data (registration is free).
- Click on the 'Data Access' tab once on the data set home page.
- Most of the RDA data is free for all users. A very few are restricted to university users or researchers. Read the individual data set home pages for usage restrictions, if any, that apply to the data set.

hardware and software

1. **HDF5**: High-performance data management and storage suite
2. **NetCDF**: 網路通用資料格式 (Network Common Data Form)
3. **compiler**(單CPU+平行化版): intel
4. **JasPer+PNG+zlib**: GRIB2

自行參閱網路資訊：

https://home.chpc.utah.edu/~u0631741/wrf-sfire/Compiling_WRF_Sfire.pdf

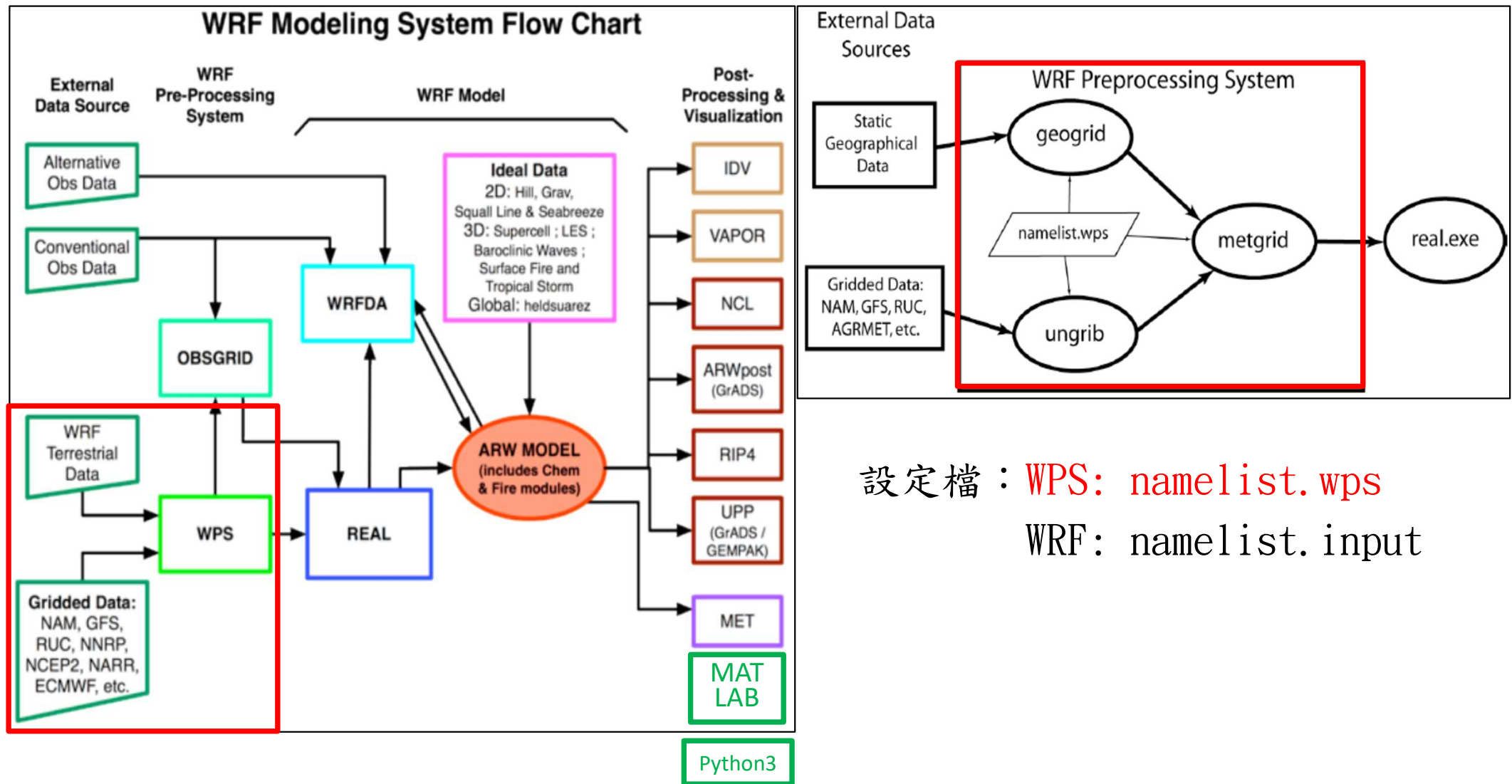
<https://blog.onemid.net/blog/wrf-install/>

不同款式硬體、不同作業環境版本常需不同版本軟體

不同款式硬體、不同作業環境版本、不同版本軟體、不同設定 = 不同模擬結果！

WRF model: WPS

1. <https://www.mmm.ucar.edu/weather-research-and-forecasting-model>



設定檔：WPS: `namelist.wps`

WRF: `namelist.input`

WPS: namelist.wps

&share [共用]

.....

/

&geogrid [地形、地表、模擬區域]

.....

/

&ungrib [grib格式全球網格資料解壓縮]

.....

/

&metgrid [整合前述資料]

.....

/

WPS: namelist.wps [共用]

&share

wrf_core = 'ARW',

max_dom = 4,

start_date = 'start_time', 'start_time', 'start_time', 'start_time',

end_date = 'end_time', 'end_time', 'end_time', 'end_time',

interval_seconds = 21600

io_form_geogrid = 2,

/

時間格式：2021-12-22_12:00:00

io_form_geogrid格式：1=binary; 2=NetCDF; 3=GRIB1

WPS: namelist.wps [地形、地表、模擬區域]

&geogrid

parent_id = 1, 1, 2, 3,

parent_grid_ratio = 1, 3, 3, 3,

i_parent_start = 1, 24, 32, 80,

j_parent_start = 1, 23, 35, 57,

s_we = 1, 1, 1, 1,

e_we = 91, 166, 223, 223,

s_sn = 1, 1, 1, 1,

e_sn = 91, 169, 223, 223,

geog_data_res = 'default', 'default', 'default', 'default',

dx = 81000,

dy = 81000,

map_proj = 'lambert',

ref_lat = 25.0,

ref_lon = 120.0,

truelat1 = 10.0,

truelat2 = 40.0,

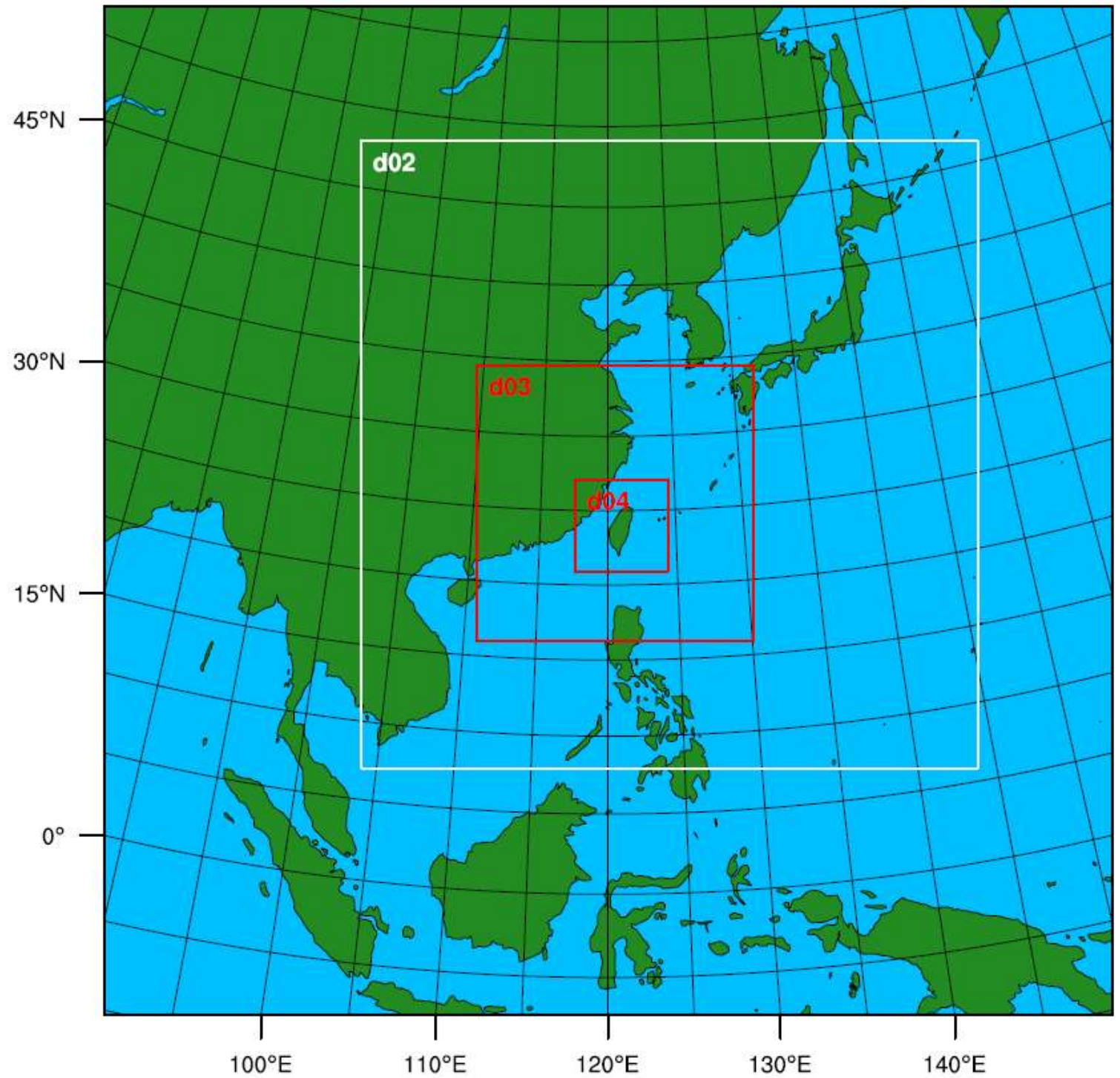
stand_lon = 120.0,

geog_data_path = '/work1/polin3916/geog',

/

WPS: geogrid

1. namelist.wps



WPS: ungrib

1. <https://rda.ucar.edu/datasets/ds083.3/#description>

[Hello Guest](#) [Register Now](#) [Sign In](#) | [Forgot Password?](#)

NCAR
UCAR

 **Research Data Archive**
Computational & Information Systems Lab

NCAR is sponsored by
National Science Foundation 

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[Home](#) [Find Data](#) [Ancillary Services](#) [About/Contact](#) [Data Citation](#) [Web Services](#) [Metrics](#) [For Staff](#)

 **NCEP GDAS/FNL 0.25 Degree Global Tropospheric Analyses and Forecast Grids**
ds083.3 | DOI: 10.5065/D65Q4T4Z

For help with this dataset, send us a message.

[Description](#) [Data Access](#) [Documentation](#) [Software](#) [Metrics](#)

Help with this page: [RDA dataset description page video tour](#)

Data Citations: [?](#) This dataset has been cited **49** times.
[▶ Published works that cited this dataset](#)

Abstract: These NCEP FNL (Final) operational global analysis and forecast data are on 0.25-degree by 0.25-degree grids prepared operationally every six hours. This product is from the Global Data Assimilation System (GDAS), which continuously collects observational data from the Global Telecommunications System (GTS), and other sources, for many analyses. The FNLs are made with the same model which NCEP uses in the Global Forecast System (GFS) but the FNLs are

Temporal Range: 2015-07-08 00:00 +0000 to 2021-12-20 15:00 +0000 (Entire dataset)
[▶ Period details by dataset product](#)

WPS: namelist. wps

&ungrib

```
  out_format = 'WPS',  
  prefix = 'FILE',  
/
```

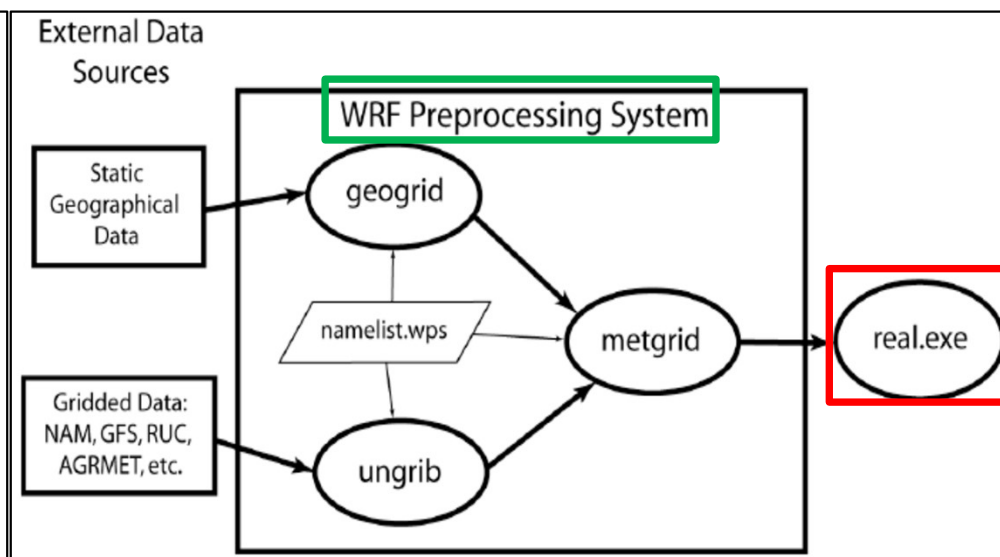
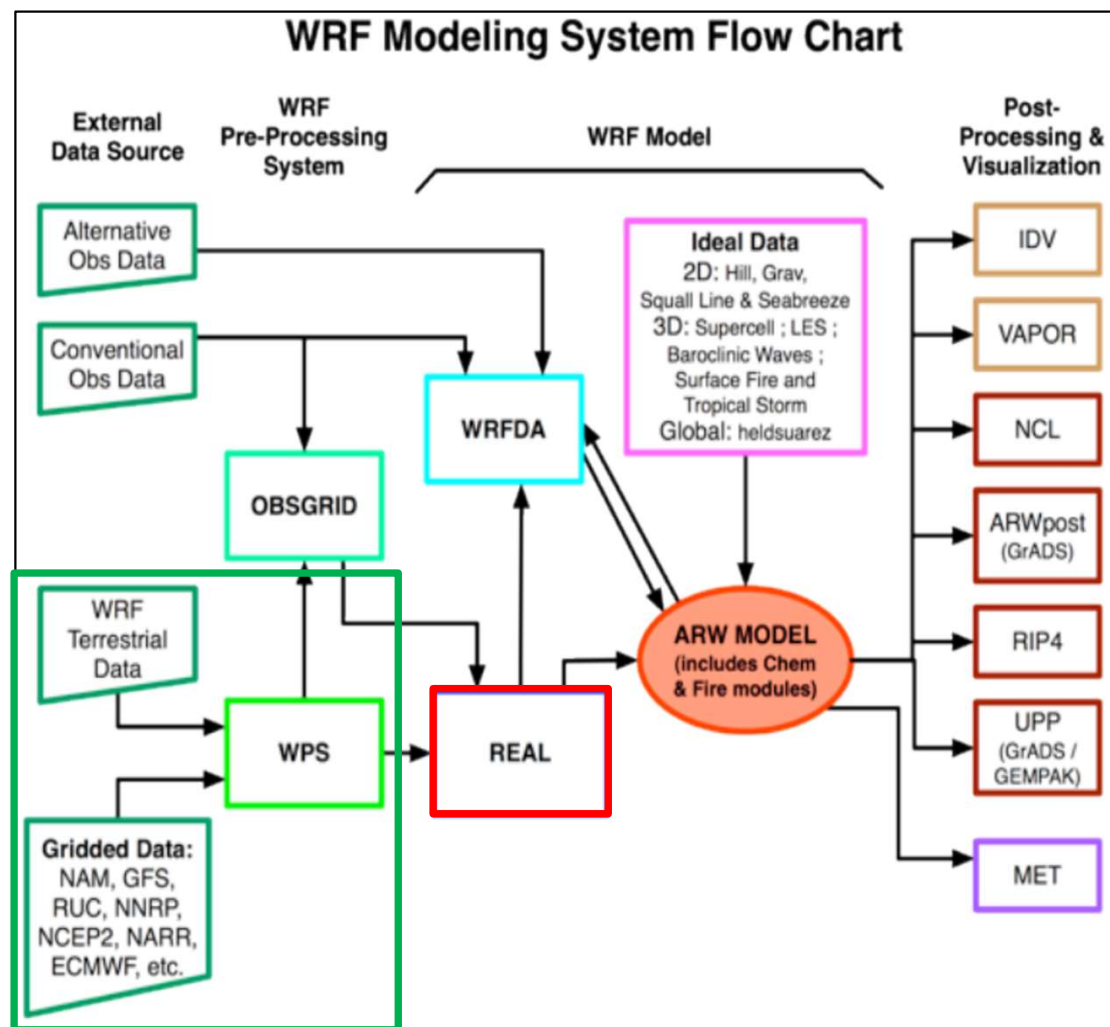
&metgrid

```
  fg_name = 'FILE'  
  io_form_metgrid = 2,  
/
```

io_form_metgrid格式：1=binary; 2=NetCDF; 3=GRIB1

WPS: real.exe

1. <https://www.mmm.ucar.edu/weather-research-and-forecasting-model>



設定檔：WPS: namelist.wps

WRF: namelist.input

real.exe 需 metgrid 完成之檔案
met_em*

單CPU: geogrid需時30分鐘(僅執行一次即可)

ungrib 3~5分鐘

metgrid 3~5分鐘

WRF: namelist.input

&time_control [時間設定]

/

&domains [範圍設定]

/

&physics [氣象物理過程設定]

/

&fdda [四維資料同化設定]

/

&dynamics [模式數值計算設定]

/

&bdy_control

/

&grib2

/

&namelist_quilt

/

在 WRF/run 子目錄下 README.namelist 簡述各設定

WRF: namelist.input [時間設定]

&time_control

```
run_days                = 1,  
run_hours               = 12, [run_minutes, run_seconds]  
start_year, start_month, start_day, start_hour, start_minute, start_second  
end_year,   end_month,   end_day,   end_hour,   end_minute,   end_second  
interval_seconds       = 21600  
input_from_file        = .true., .true., .true., .true.,  
history_interval       = 60,    60,    60,    60,  
frames_per_outfile     = 1000, 1000, 1000, 1000,  
restart                = .false.,  
restart_interval       = 5000,  
io_form_history        = 2  
io_form_restart        = 2  
io_form_input          = 2  
io_form_boundary       = 2  
debug_level            = 0  
/
```

WRF: namelist.input [範圍設定]

```
&domains
time_step                = 240,
max_dom                  = 4,
e_we                    = 91,      166,      223,      223,
e_sn                    = 91,      169,      223,      223,
e_vert                  = 25,      25,      25,      25,
p_top_requested          = 10000,
num_metgrid_levels       = 32,
num_metgrid_soil_levels  = 4,
dx                      = 81000,  27000,  9000,   3000,
dy                      = 81000,  27000,  9000,   3000,
grid_id                  = 1,      2,      3,      4,
parent_id                = 0,      1,      2,      3,
i_parent_start           = 1,      24,     32,     80,
j_parent_start           = 1,      23,     35,     57,
parent_grid_ratio         = 1,      3,      3,      3,
parent_time_step_ratio   = 1,      3,      3,      3,
feedback                 = 0,
smooth_option            = 0
eta_levels                = 1.000,  0.998,  0.996,  0.994,  0.992,
                        .....
                        0.100,  0.070,  0.040,  0.020,  0.0,

/
```

num_metgrid_levels: (2016 4/26 之前)27, (2019 6/11 之前)32, 之後34

time step for integration in integer seconds recommend $6 \times dx$ (in km) for typical real-data cases [$6 \times 81\text{km} = 486\text{s}$] => CFL condition

WRF: namelist.input [氣象物理過程設定]

&physics

mp_physics	= 18,	18,	18,	18,	[NSSL 2-moment]
ra_lw_physics	= 5,	5,	5,	5,	[Goddard]
ra_sw_physics	= 2,	2,	2,	2,	[Goddard]
radt	= 25,				
sf_sfclay_physics	= 5,	5,	5,	5,	[MYNN]
sf_surface_physics	= 2,	2,	2,	2,	[Unified Noah]
bl_pbl_physics	= 5,	5,	5,	5,	[MYNN2.5]
bldt	= 0,	0,	0,	0,	
cu_physics	= 3,	3,	3,	0,	[Grell-Freitas]
cudt	= 0,				
isfflx	= 1,				
ifsnow	= 1,				
icloud	= 1,				
surface_input_source	= 3,				
num_soil_layers	= 4,				
num_land_cat	= 21,				
sf_urban_physics	= 0,				
/					

WRF: namelist.input

&fdda

/

&dynamics

w_damping

diff_opt

km_opt

diff_6th_opt

diff_6th_factor

base_temp

damp_opt

zdamp

dampcoef

khdif

kvdif

non_hydrostatic

moist_adv_opt

scalar_adv_opt

gwd_opt

/

&bdy_control

spec_bdy_width

spec_zone

relax_zone

specified

nested

/

&grib2

/

&namelist_quilt

nio_tasks_per_group = 0,

nio_groups = 1, /

/

```
= 0,  
= 1,      1,      1,      1,  
= 4,      4,      4,      4,  
= 0,      0,      0,      0,  
= 0.12,   0.12,   0.12,   0.12,  
= 290.  
= 0,  
= 5000.,  5000.,  5000.,  5000.,  
= 0.2,    0.2,    0.2,    0.2,  
= 0,      0,      0,      0,  
= 0,      0,      0,      0,  
= .true., .true., .true., .true.,  
= 1,      1,      1,      1,  
= 1,      1,      1,      1,  
= 1,
```

```
= 5,  
= 1,  
= 4,  
= .true., .false.,.false.,.false.,  
= .false., .true., .true., .true.,
```

WRF executed on **Taiwanial** at NCHC

1. 申請國網帳號：網頁帳號+網頁密碼+Twn1帳號+Twn1(每30秒變更)密碼
2. 安裝相關軟體：**HDF5**、**NetCDF**、**compiler**、**JasPer**+**PNG**+**zlib**、**WPS**、**WRF**
3. **Portable Batch System** 排程(queue)設定檔：

```
#!/bin/bash
```

```
#PBS -l select=5:ncpus=40:mpiprocs=40 [CPU core 數=5×40]
```

```
#PBS -N WRF0401
```

```
#PBS -q ct400 [CPU群組]
```

```
#PBS -P XXXXXXXXXX [扣款計劃編號]
```

```
#PBS -j oe
```

```
cd $PBS_O_WORKDIR
```

```
module load intel/2018_u1
```

```
export I_MPI_HYDRA_PMI_CONNECT=alltoall
```

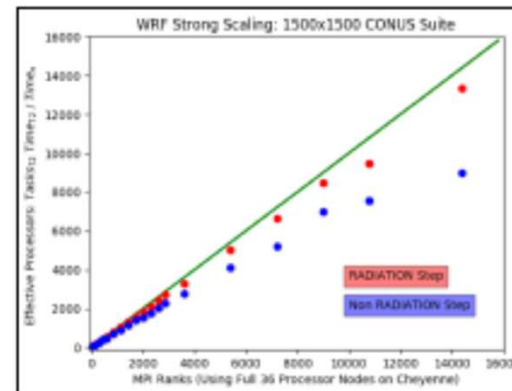
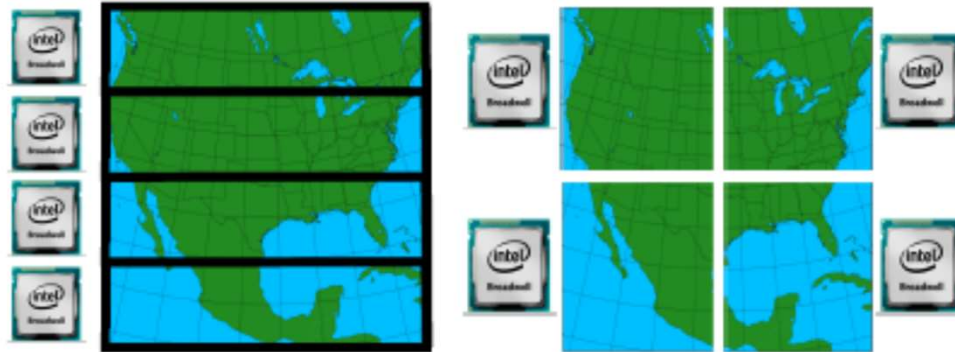
```
export I_MPI_HYDRA_BRANCH_COUNT=-1
```

```
mpiexec.hydra -PSM2 ./wrf.exe
```

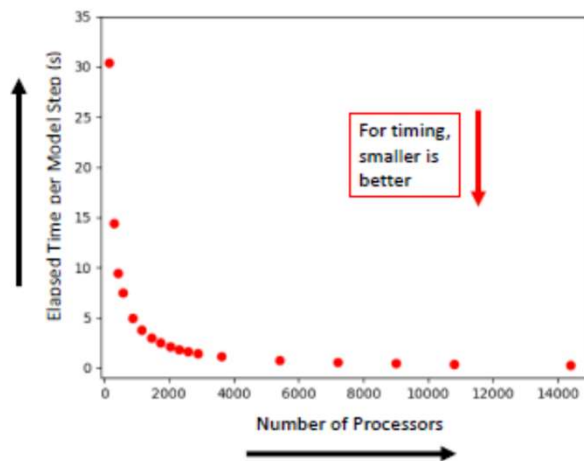
國網有提供相關說明示範 pdf 檔案

WRF on super-computer (Presentation of Dave Gill @NCAR)

- Parallelism in WRF
 - Why is it required
 - What is “scaling”
- OpenMP and MPI
- Halos
- Domain Decomposition
 - Reasonable, max, min, bad
 - Dimensions and decomposition
 - Coarse grid vs nest grid
- Where is all of this information
 - Using rsl files

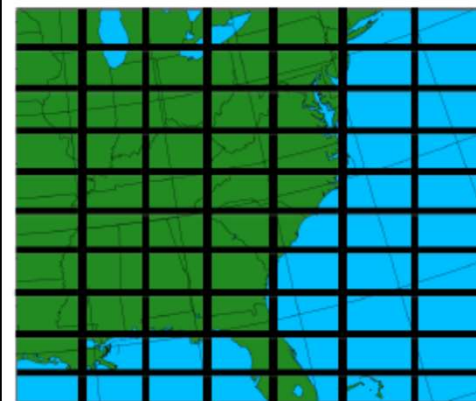


What is “scaling”



- While the information is accurate, this presentation of the timing data is not easy to interpret
- Scaling plots are typically recast with an easy comparison against a known value (100% efficient) or a fixed quantity (such as Days of Simulation per Elapsed Day of Computation)

- 70 tasks
- 10 (j) x 7 (i)



- 71 tasks
- 71 (j) x 1 (i)



Minimum decomposed computational patch size, either x-dir or y-dir, is 10 grid cells.

案例

1. 環保署-成大計劃 2019年 WRF 與氣象局地面測站校驗
default: Goddard (可考慮改為 RRTMG), MYNN2.5
2. 風能風塔高度 2016年 WRF 風場模擬
模擬策略規劃
3. 環保署-雲科大計劃 2018年 WRF 模擬 PBLH
模擬策略規劃
4. 不同 CPU 數測試模擬
.....

5月PBL scheme: | ACM2 | - | MYNN | , 各分區、各標準, 各內插統計差異

5月PBL scheme: | ACM2 | - | MYNN | , 各分區、各標準, 各內插統計差異

MYNN較好

區域四點	dt	abs(dt)	dws	sqrt(dws^2)	dwd	abs(dwd)	dt>60%	abs(dt)>60%	dws>60%	sqrt(dws^2)>60%	dwd>60%	abs(dwd)>60%
台灣地區	0.370902948	0.078196891	0.524466022	0.287255164	0.004387183	-0.012162295	-0.083333333	0	-0.041666667	-0.166666667	0.041666667	0.041666667
台灣地區	0	0	0	0	0	0	0	0	0	0	0	0
北部地區	0.311756766	0.079960925	0.452798042	0.362564489	-0.001357614	0.001527044	0	0	-0.1	-0.2	0	0
北部地區	0	0	-1	0	0	0	0	0	-1	0	0	0
中部地區	0.305303865	0.16448177	0.773378922	-0.040532278	-0.0000152348	-0.059065548	-0.333333333	0	0	0.166666667	0.166666667	0.166666667
中部地區	0	0	0	0	0	0	0	0	0	0	0	0
雲嘉南地區	0.262033798	0.166208109	0.591088382	-0.050558156	-0.001952071	-0.046007173	-0.142857143	0	0	0.142857143	0.142857143	0.142857143
雲嘉南地區	0	0	0	0	0	0	0	0	0	0	0	0
南部地區	0.425565621	0.081965336	0.421768398	0.025065039	-0.003279993	-0.027855673	0	0	0.285714286	-0.142857143	0	0.142857143
南部地區	0	0	0	0	0	0	0	0	1	0	0	0
東部地區	0.542267553	0.138759727	1.055226543	0.300563494	0.009458933	-0.0437111	-0.125	0	-0.125	-0.125	0.125	0.125
東部地區	0	0	0	0	0	0	-1	0	-1	0	0	0
最近點	0	0	0	0	0	0	0	0	0	0	0	0
台灣地區	0.366946165	0.018859193	0.569905415	0.280147679	0.002605399	-0.015437912	0	0	-0.125	-0.041666667	0.041666667	0.041666667
台灣地區	0	0	0	0	0	0	0	0	-1	0	0	0
北部地區	0.288867838	0.045276022	0.467005052	0.364560471	0.002104694	-1.65425E-05	0.1	0	-0.2	-0.1	0	0
北部地區	0	0	-1	0	0	0	0	0	-1	0	0	0
中部地區	0.267586593	0.034398271	0.912205786	0.011297363	-0.011655209	-0.059954863	-0.166666667	0	0	0.166666667	0.166666667	0.166666667
中部地區	-1	0	0	0	0	0	0	0	0	0	0	0
雲嘉南地區	0.236695204	0.055696594	0.709108987	-0.00028425	-0.012790975	-0.046863186	-0.142857143	0	0	0.142857143	0.142857143	0.142857143
雲嘉南地區	-1	0	0	0	0	0	0	0	0	0	0	0
南部地區	0.458255445	-0.023627085	0.557457053	-0.007464847	0.001163004	-0.030390299	-0.142857143	0	0.142857143	0.142857143	0	0.142857143
南部地區	0	0	0	0	0	0	-1	0	1	0	0	0
東部地區	0.534663908	0.018292206	1.157199349	0.262715112	0.021076382	-0.050923256	-0.125	0	-0.125	0	0.125	0.125
東部地區	0	0	0	0	0	0	0	0	0	0	0	0
陸地點	0	0	0	0	0	0	0	0	0	0	0	0
台灣地區	0.345267403	0.06269232	0.482152675	0.228340923	0.002098185	-0.014295336	-0.041666667	-0.041666667	-0.041666667	-0.125	0.041666667	0.041666667
台灣地區	0	0	0	0	0	0	0	0	0	0	0	0
北部地區	0.319204257	0.071735068	0.454468448	0.36464016	-0.001032065	0.001743249	0	0	-0.1	-0.3	0	0
北部地區	0	0	-1	0	0	0	0	0	-1	0	0	0
中部地區	0.319247917	0.198582669	0.688593619	-0.003408518	-0.009524528	-0.061766768	-0.166666667	-0.166666667	0.166666667	0.166666667	0.166666667	0.166666667
中部地區	0	0	0	0	0	0	0	0	0	0	0	0
雲嘉南地區	0.273808653	0.194621865	0.520610478	-0.0165049	-0.006607406	-0.048310313	-0.142857143	-0.142857143	0.142857143	0.142857143	0.142857143	0.142857143
雲嘉南地區	0	0	0	0	0	0	0	0	0	0	0	0
南部地區	0.368466423	0.093807331	0.324928057	-0.042466119	-0.011473988	-0.030886095	0	-0.142857143	0.142857143	0	0.142857143	0.142857143
南部地區	0	0	0	0	0	0	0	0	0	0	0	0
東部地區	0.503559395	0.077678513	0.939962627	0.112566954	0.003800733	-0.050928164	-0.125	-0.125	0	0.125	0.125	0.125
東部地區	0	0	0	0	0	0	-1	0	0	0	0	0
ACM2 MYNN ACM2-MYNN ⊕												

1. 2019 WRF校驗結果

5月PBL scheme: | YSU | - | MYNN | , 各分區、各標準, 各內插統計差異

YSU較好

MYNN較好

區域	dt	abs(dt)	dws	sqrt(dws^2)	dwd	abs(dwd)	dt>60%	abs(dt)>60%	dws>60%	sqrt(dws^2)>60%	dwd>60%	abs(dwd)>60%
四點												
台灣地區	0.115722887	-0.001029758	0.123416279	0.096832308	0.002561819	0.002982608	0	0	-0.041666667	-0.083333333	0	0
台灣地區	0	0	0	0	0	0	0	0	0	0	0	0
北部地區	0.037776689	-0.022894668	0.125917803	0.094252487	-0.003257104	0.001705349	0	0	-0.1	-0.1	0	0
北部地區	0	0	0	0	0	0	0	0	-1	0	0	0
中部地區	0.202840707	0.057646001	0.08733964	-0.094281926	0.004758058	0.004843338	0	0	0	0.166666667	0	0
中部地區	0	0	0	0	0	0	0	0	0	0	0	0
雲嘉南地區	0.159990186	0.04943493	0.016634512	-0.10061412	-0.00359295	0.00649625	0	0	0	0.142857143	0	0
雲嘉南地區	0	0	0	0	0	0	0	0	0	0	0	0
南部地區	0.147025344	0.033282947	0.150248014	0.133486299	0.004594228	0.006150295	0	0	0	-0.142857143	-0.142857143	0
南部地區	0	0	0	0	0	0	0	0	0	0	0	0
東部地區	0.291425539	0.055962025	0.158408984	0.045721981	0.005629164	0.005057419	0	0	0	0	0	0
東部地區	0	0	0	0	0	0	0	0	0	0	0	0
最近點	0	0	0	0	0	0	0	0	0	0	0	0
台灣地區	0.124461719	-0.055548405	0.132286537	0.107605371	-0.001194218	0.003453849	0.041666667	0	-0.083333333	-0.041666667	-0.041666667	0
台灣地區	0	0	0	0	0	0	0	0	0	0	0	0
北部地區	0.020905779	-0.052865445	0.132162182	0.11547161	-0.000198782	0.002290849	0.1	0	-0.2	0	0	0
北部地區	0	0	0	0	0	0	0	0	-1	0	0	0
中部地區	0.20975062	-0.039502987	0.09318087	-0.106536721	-0.011087107	0.004753397	0	0	0	0	-0.166666667	0
中部地區	-1	0	0	0	0	0	0	0	0	0	0	0
雲嘉南地區	0.166812776	-0.027415033	0.037419023	-0.09927125	-0.00832756	0.006198659	0	0	0	0	-0.142857143	0
雲嘉南地區	-1	0	0	0	0	0	0	0	0	0	0	0
南部地區	0.176255388	-0.064531195	0.163435143	0.132649961	0.009049135	0.006149159	0	0	0	0	-0.142857143	0
南部地區	0	0	0	0	0	0	0	0	0	0	0	0
東部地區	0.313766704	-0.042658074	0.17416918	0.031185808	0.010819432	0.006156965	0	0	0	-0.125	-0.125	0
東部地區	0	0	0	0	0	0	0	0	0	0	0	0
陸地點	0	0	0	0	0	0	0	0	0	0	0	0
台灣地區	0.082086745	-0.015262984	0.108293544	0.063089325	0.002415739	0.002845236	0	-0.041666667	-0.041666667	-0.083333333	0	0
台灣地區	0	0	0	0	0	0	0	0	0	0	0	0
北部地區	0.048748797	-0.0239396	0.124192398	0.090402603	-0.003251811	0.001511322	0	0	-0.1	-0.2	0	0
北部地區	0	0	0	0	0	0	0	0	-1	0	0	0
中部地區	0.190935809	0.064161786	0.078219064	-0.103875887	0.006018694	0.004178378	0	-0.166666667	0	0.166666667	0	0
中部地區	0	0	0	0	0	0	0	0	0	0	0	0
雲嘉南地區	0.15036785	0.056457294	0.012082097	-0.106089495	0.003622854	0.006018312	-0.142857143	-0.142857143	0	0.142857143	0	0
雲嘉南地區	0	0	0	0	0	0	0	0	0	0	0	0
南部地區	0.07685962	0.027056269	0.117511206	0.080376122	0.003909766	0.005067001	0	-0.142857143	0	0	0	0
南部地區	0	0	0	0	0	0	0	0	0	0	0	0
東部地區	0.223537367	0.002119607	0.137527107	-0.034472665	0.005667189	0.004513197	-0.125	-0.125	0	0.125	0	0
東部地區	0	0	0	0	0	0	-1	0	0	0	0	0

1. 2019 WRF校驗結果 (RRTMG計算時間多1倍)

5月RAD. scheme: **RRTMG** – **Goddard** , 各分區、各標準, 各內插統計差異

RRTMG較好

Goddard較好

區域	dt	abs(dt)	dws	sqrt(dws^2)	dwd	abs(dwd)	dt>60%	abs(dt)>60%	dws>60%	sqrt(dws^2)>60%	dwd>60%	abs(dwd)>60%
四點												
台灣地區	0.050344542	-0.035471128	-0.070894894	-0.069891295	-0.000482928	-0.002613443	-0.041666667	0	0.041667	0	0	0
台灣地區	0	0	0	0	0	0	0	0	0	0	0	0
北部地區	0.01259541	-0.010389652	-0.088452236	-0.069682139	-0.000433668	-0.003632841	0	0	0	0	0	0
北部地區	0	0	0	0	0	0	0	0	0	0	0	0
中部地區	-0.038445163	-0.062171301	-0.025951456	-0.038283555	-0.000378838	0.002688866	-0.166666667	0	0	0	0	0
中部地區	0	0	0	0	0	0	0	0	0	0	0	0
雲嘉南地區	-0.013569584	-0.035151175	-0.031084451	-0.047191293	-0.004030276	0.002694298	0	0	0	0	0	0
雲嘉南地區	0	0	0	0	0	0	0	0	0	0	0	0
南部地區	0.114114594	-0.025295405	-0.07944721	-0.075737891	0.0040186	-0.001775324	0	0.142857	0	0	0	0
南部地區	0	0	0	0	0	0	0	0	0	0	0	0
東部地區	0.052830618	-0.084757558	-0.037315243	-0.081632286	-0.007786694	-0.003419589	0	0	0	0	0	0
東部地區	0	0	0	0	0	0	0	0	0	0	0	0
最近點												
台灣地區	0.046328292	-0.039719291	-0.072319709	-0.068966185	-0.000204264	-0.002936059	0.083333333	0	0	0	0	0
台灣地區	0	0	0	0	0	0	0	0	0	0	0	0
北部地區	0.022585525	-0.017237504	-0.090874333	-0.074502371	-0.001014524	-0.003854435	0.1	0	0	0	0	0
北部地區	0	0	0	0	0	0	0	0	0	0	0	0
中部地區	-0.059547772	-0.063463344	-0.032692012	-0.01905361	-0.000978973	0.002941089	0.166666667	0	0	0	0	0
中部地區	0	0	0	0	0	0	0	0	0	0	0	0
雲嘉南地區	-0.02624763	-0.037701861	-0.036453927	-0.036823689	-0.002896337	0.002610919	0.142857143	0	0	0	0	0
雲嘉南地區	0	0	0	0	0	0	0	0	0	0	0	0
南部地區	0.102457577	-0.033926577	-0.079927773	-0.060662818	0.003192278	-0.002640913	0	0	0	0	0	0
南部地區	0	0	0	0	0	0	0	0	0	0	0	0
東部地區	0.019535764	-0.076383197	-0.043797523	-0.075355138	-0.004687409	-0.003541599	0.125	0	0	0	0	0
東部地區	0	0	0	0	0	0	0	0	0	0	0	0
陸地點												
台灣地區	0.024875604	-0.020798836	-0.067740044	-0.065606922	-0.000613667	-0.00300557	0	0	0	0.041666667	0	0
台灣地區	0	0	0	0	0	0	0	0	0	0	0	0
北部地區	0.010112755	-0.009849994	-0.088178637	-0.06916609	0.000258048	-0.003623175	0	0	0	0	0	0
北部地區	0	0	0	0	0	0	0	0	0	0	0	0
中部地區	-0.040755764	-0.057950718	-0.022073113	-0.042890348	0.000579215	0.001406171	0	0	0	0.166666667	0	0
中部地區	0	0	0	0	0	0	0	0	0	0	0	0
雲嘉南地區	-0.016402192	-0.031855345	-0.028554553	-0.052139153	0.002949315	0.001665424	0	0	0	0.142857143	0	0
雲嘉南地區	0	0	0	0	0	0	0	0	0	0	0	0
南部地區	0.069068305	0.008897983	-0.08049176	-0.071165995	0.003458719	-0.002704603	0	0	0	0.142857143	0	0
南部地區	0	0	0	0	0	0	0	0	0	0	0	0
東部地區	-0.002999297	-0.060811726	-0.029538135	-0.076192649	-0.007025841	-0.004894611	0	0	0	0.125	0	0
東部地區	0	0	0	0	0	0	0	0	0	0	0	0

1. 2019 WRF校驗結果 (RRTMG計算時間多1倍)

7月RAD. scheme: **RRTMG** – **Goddard** , 各分區、各標準, 各內插統計差異

RRTMG較好

Goddard較好

區域	dt	abs(dt)	dws	sqrt(dws^2)	dwd	abs(dwd)	dt>60%	abs(dt)>60%	dws>60%	sqrt(dws^2)>60%	dwd>60%	abs(dwd)>60%
四點												
台灣地區	0.007891061	-0.047384423	-0.041819366	-0.050216676	-0.001844319	-0.007862159	0	0	0.083333333	0	0	0
台灣地區	0	0	0	0	0	0	0	0	0	0	0	0
北部地區	0.001551724	-0.01792474	-0.012209726	-0.028070108	0.005374197	-0.008934689	0	0	0	0	0	0
北部地區	0	0	0	0	0	0	0	0	0	0	0	0
中部地區	-0.03761171	-0.068079635	0.009832979	-0.043856392	-0.001486597	-0.004495961	0	0	0.166666667	0	0	0
中部地區	0	0	0	0	0	0	0	0	0	0	0	0
雲嘉南地區	-0.034194534	-0.068856665	-0.031838855	-0.047893802	-0.001166199	-0.005863644	0	0	0.142857143	0	0	0
雲嘉南地區	0	0	0	0	0	0	0	0	0	0	0	0
南部地區	0.033768921	-0.061231407	-0.089839484	-0.080727453	-0.002569841	-0.007390625	0	0	0.142857143	0	0	0
南部地區	0	0	0	0	0	0	0	0	0	0	0	0
東部地區	-0.014431315	-0.083028144	-0.030173859	-0.048351555	-0.001847564	-0.006580471	0	0	0.125	0	0	0
東部地區	0	0	0	0	0	0	0	0	0	0	0	0
最近點	0	0	0	0	0	0	0	0	0	0	0	0
台灣地區	0.004857369	-0.040347734	-0.040320573	-0.043301792	-0.002571492	-0.007216627	0	0	0	0	0	0
台灣地區	0	0	0	0	0	0	0	0	0	0	0	0
北部地區	-0.003903171	-0.010959298	-0.013890004	-0.02489058	0.004421084	-0.007543446	0	0	0	0	0	0
北部地區	0	0	0	0	0	0	0	0	0	0	0	0
中部地區	-0.053403156	-0.06339958	0.01307267	-0.039512323	-0.002699225	-0.003366645	0	0	0	0	0	0
中部地區	0	0	0	0	0	0	0	0	0	0	0	0
雲嘉南地區	-0.041908932	-0.066398322	-0.035269094	-0.044387515	-0.002398074	-0.004939868	0	0	0	0	0	0
雲嘉南地區	0	0	0	0	0	0	0	0	0	0	0	0
南部地區	0.038568025	-0.063387297	-0.078076336	-0.06163152	-0.00522403	-0.006515571	0	0	0	0	0	0
南部地區	0	0	0	0	0	0	0	0	0	0	0	0
東部地區	0.040291709	-0.071419324	-0.022771832	-0.035693013	-0.000348749	-0.006334748	0	0	0	0	0	0
東部地區	0	0	0	0	0	0	0	0	0	0	0	0
陸地點	0	0	0	0	0	0	0	0	0	0	0	0
台灣地區	-0.004649705	-0.039511814	-0.032571182	-0.038522149	-0.001860061	-0.008040932	0	0	0	0	0	0
台灣地區	0	0	0	0	0	0	0	0	0	0	0	0
北部地區	0.004867576	-0.018135146	-0.012754051	-0.026591318	0.004807475	-0.00873722	0	0	0	0	0	0
北部地區	0	0	0	0	0	0	0	0	0	0	0	0
中部地區	-0.036675052	-0.06477183	0.007683467	-0.046091953	0.001256665	-0.004257983	0	0	0	0	0	0
中部地區	0	0	0	0	0	0	0	0	0	0	0	0
雲嘉南地區	-0.035182527	-0.067336696	-0.030118445	-0.050407434	0.001424596	-0.005614494	0	0	0	0	0	0
雲嘉南地區	0	0	0	0	0	0	0	0	0	0	0	0
南部地區	0.0072779	-0.049896623	-0.07682855	-0.062872933	0.00082718	-0.007315274	0	0	0	0	0	0
南部地區	0	0	0	0	0	0	0	0	0	0	0	0
東部地區	-0.032618663	-0.065039758	0.00497151	-0.022294533	6.20831E-05	-0.007469409	0	0	0	0	0	0
東部地區	0	0	0	0	0	0	0	0	0	0	0	0

2. 風場校驗： WRF: 2015 10/07 00:00Z~11/06 16:00Z (4417 points)
(25day+5day)*24hr+16hr=736hr, 736*6+1=4417

福海觀測資料：2015 10/06 16:10Z~11/06 16:00Z

Wind Speed

wrf L0: 10 m

wrf L1: 17.9m

wrf L2: 44.4m

wrf L3: 80.0m

wrf L4:107.0m

wrf L5:148.4m

obs. 86 m

BIAS=-0.4888

RMSE= 2.6885

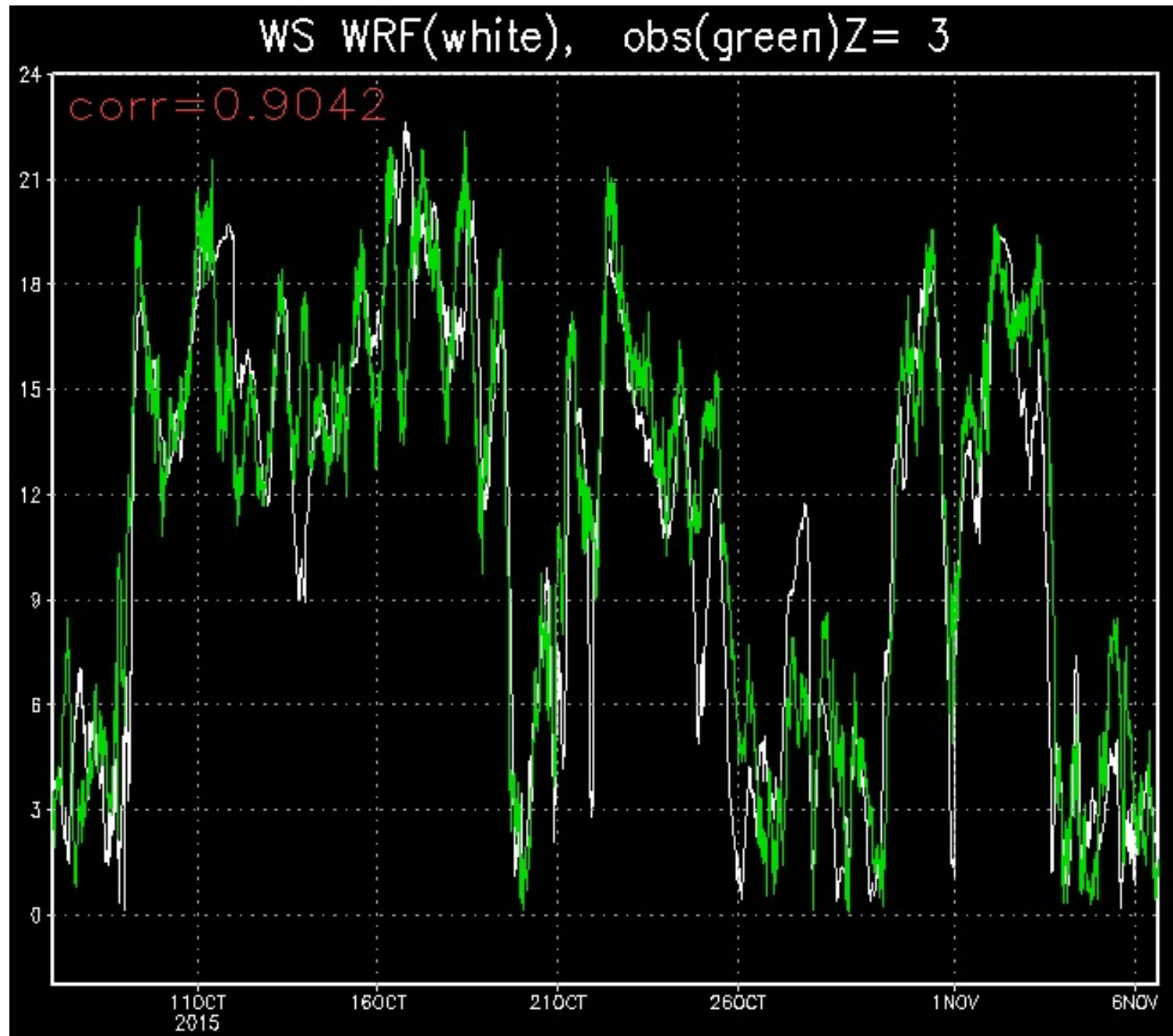
STDE= 2.6437

AVG_{WRF}=10.8698

AVG_{OBS}=11.3586

STD_{WRF}= 6.1423

STD_{OBS}= 5.8822



2. 風場校驗：2015 10/07 00:00Z~11/06 16:00Z (25day+5day)

worst corr. day

best corr. day

Wind Speed

wrf L0: 10 m

wrf L1: 17.9m

wrf L2: 44.4m

wrf L3: 80.0m

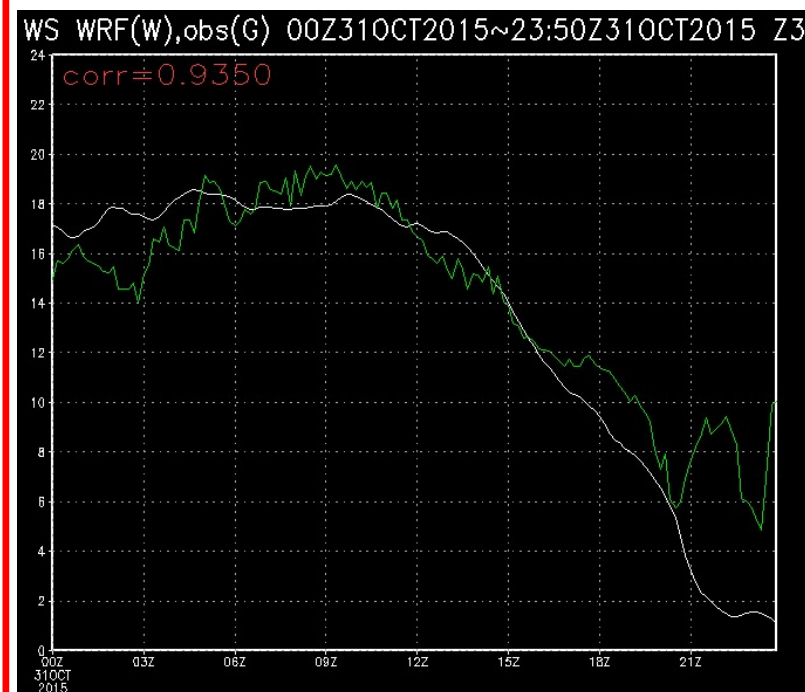
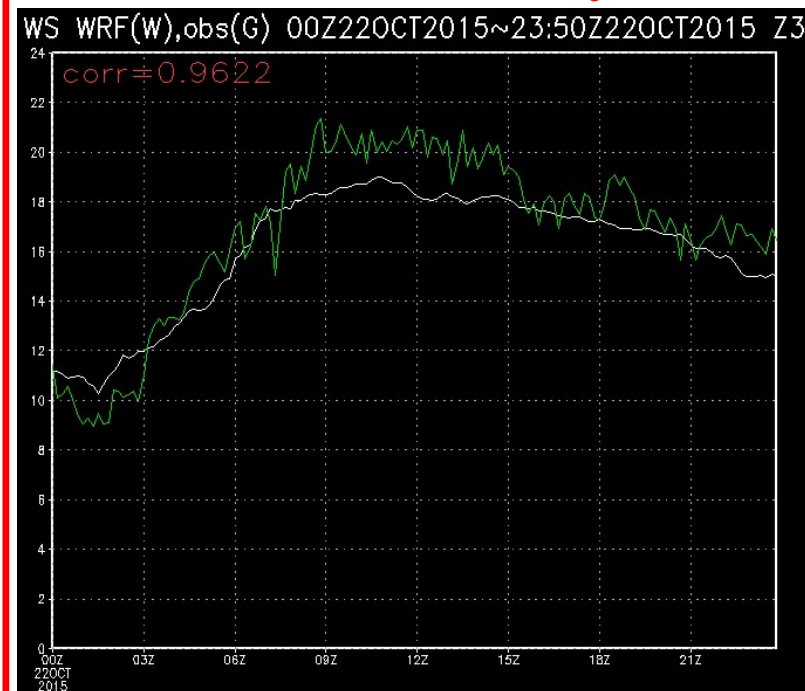
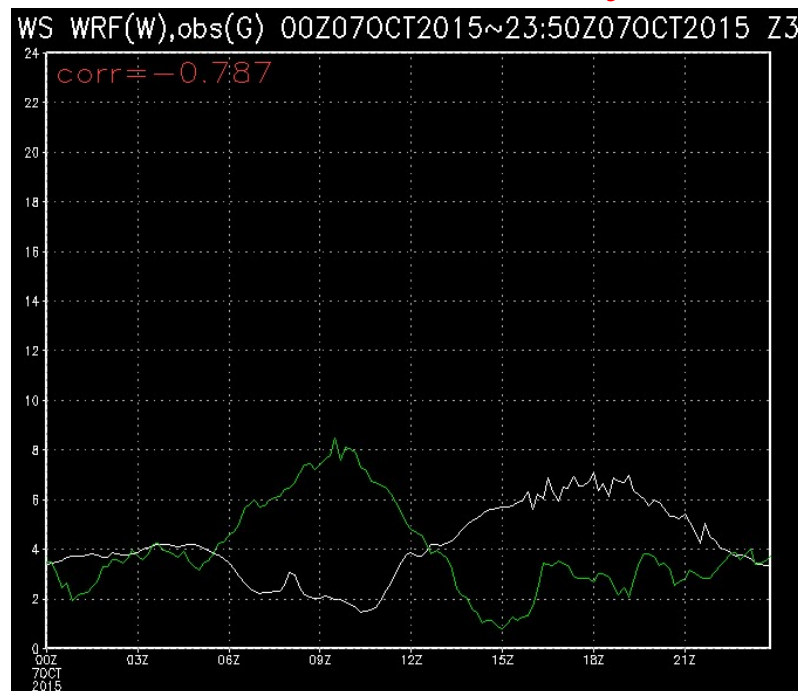
wrf L4: 107.0m

wrf L5: 148.4m

obs. 86 m

單日corr.

-0.787~0.9622



2. WRF 重疊區 (for example)

D_{-1} 12Z $\sim$ D_{+1} 00Z(白色) , D_0 12Z $\sim$ D_{+2} 00Z(紅色)

(藍色) = (白色) - (紅色)

Wind Speed

wrf L0: 10 m

wrf L1: 17.9m

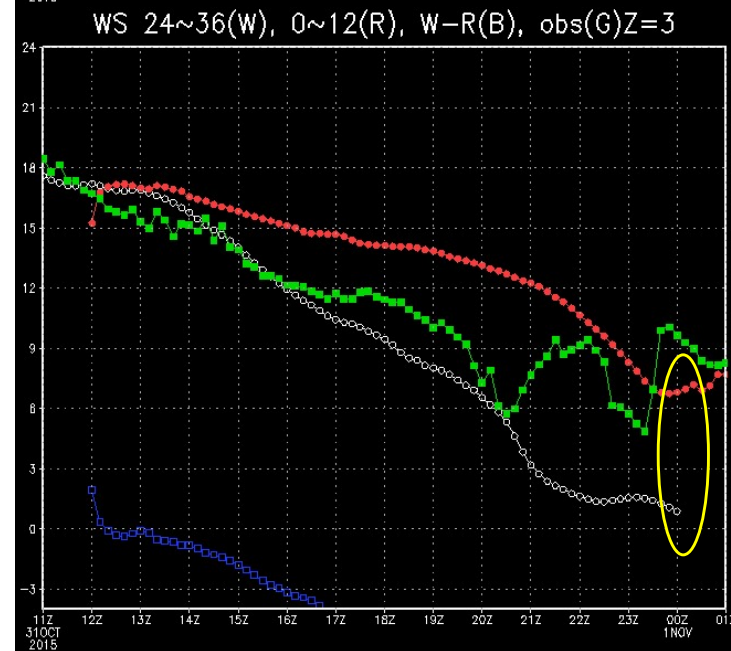
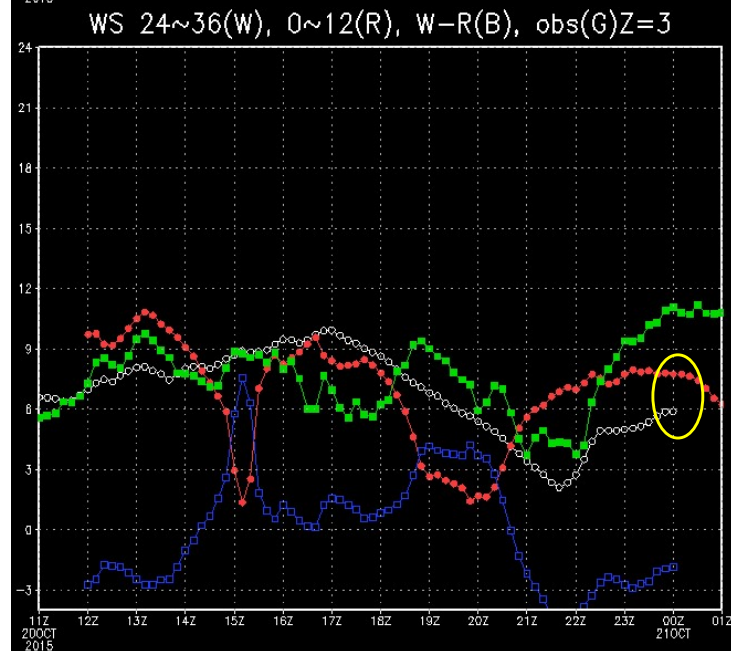
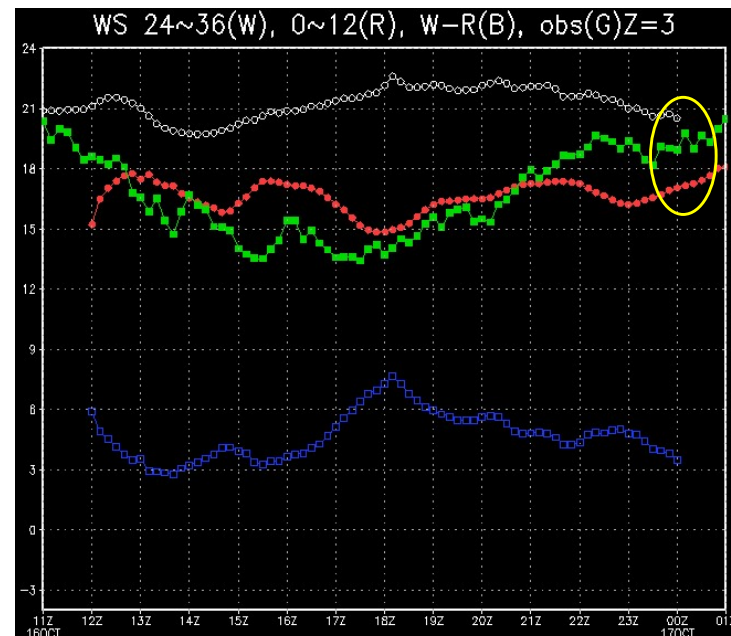
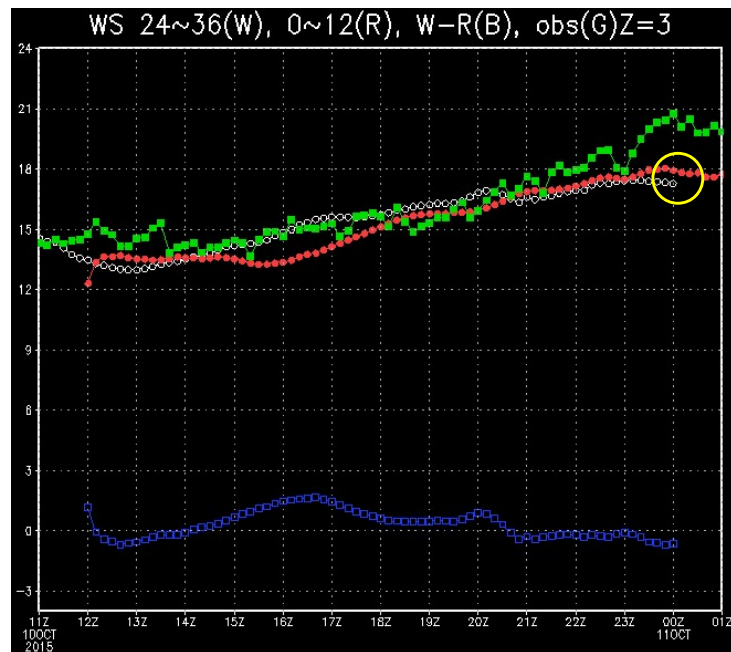
wrf L2: 44.4m

wrf L3: 80.0m

wrf L4: 107.0m

wrf L5: 148.4m

obs. 86 m



3. PBL height comparison

高雄站： 120.3157E, 22.5659N

最近網格(56, 24)=120.302E, 22.5646N, T:corr.=0.8772

4點內插

T:corr.=0.9073

(windspeed: nearP:0.5141 vs. 4P:0.5295)

3點內插 (排除海面點) :

T:corr.=0.9485

(windspeed:

3P:corr.=0.5591)

WRF data: 41 days

2018 1/1~2/9 (00-23 UTC=F12-35)

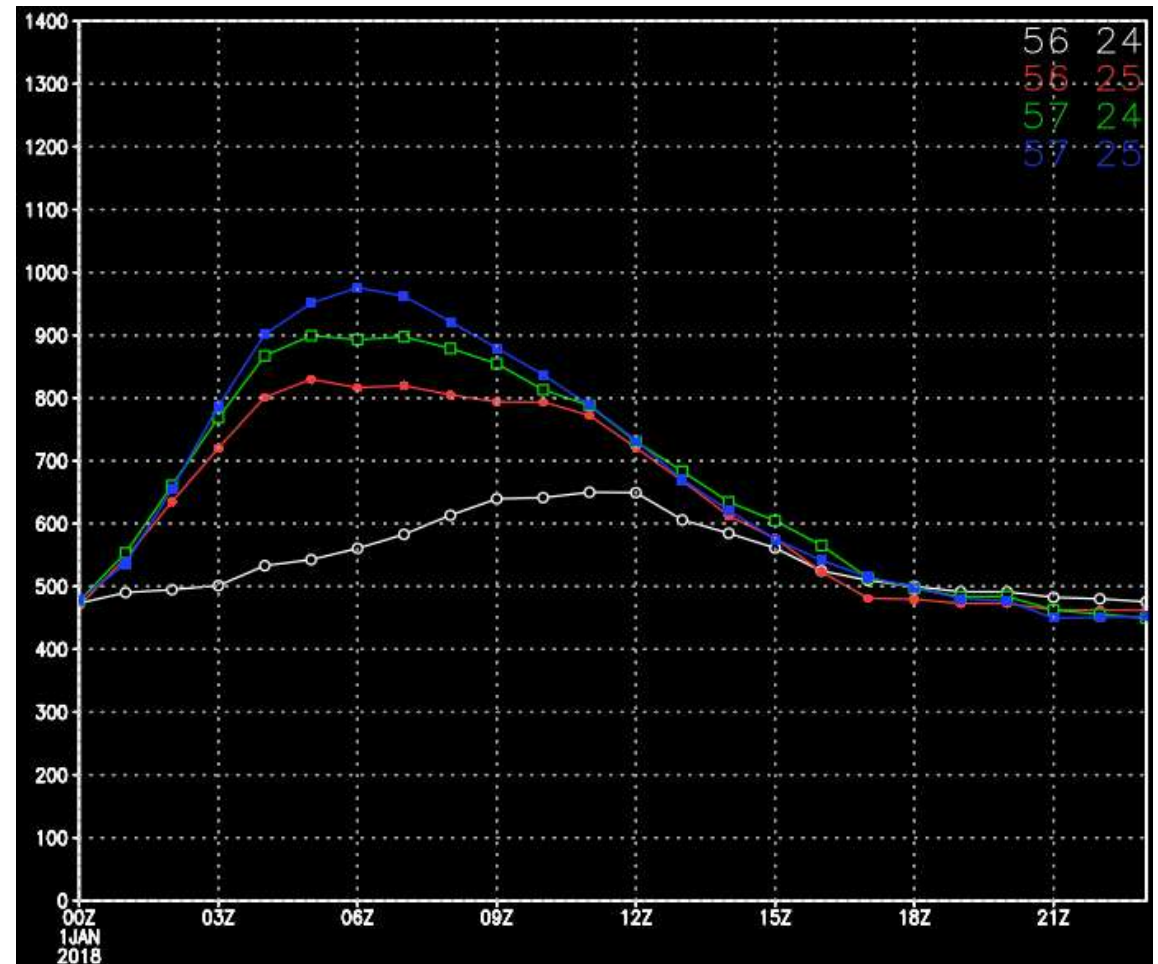
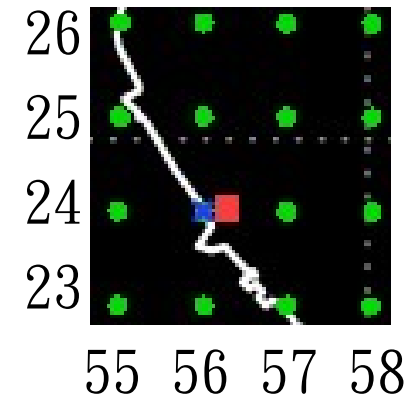
lu_index

(56, 24)=17 'Water'

(57, 24)=13 'Urban and Built-Up'

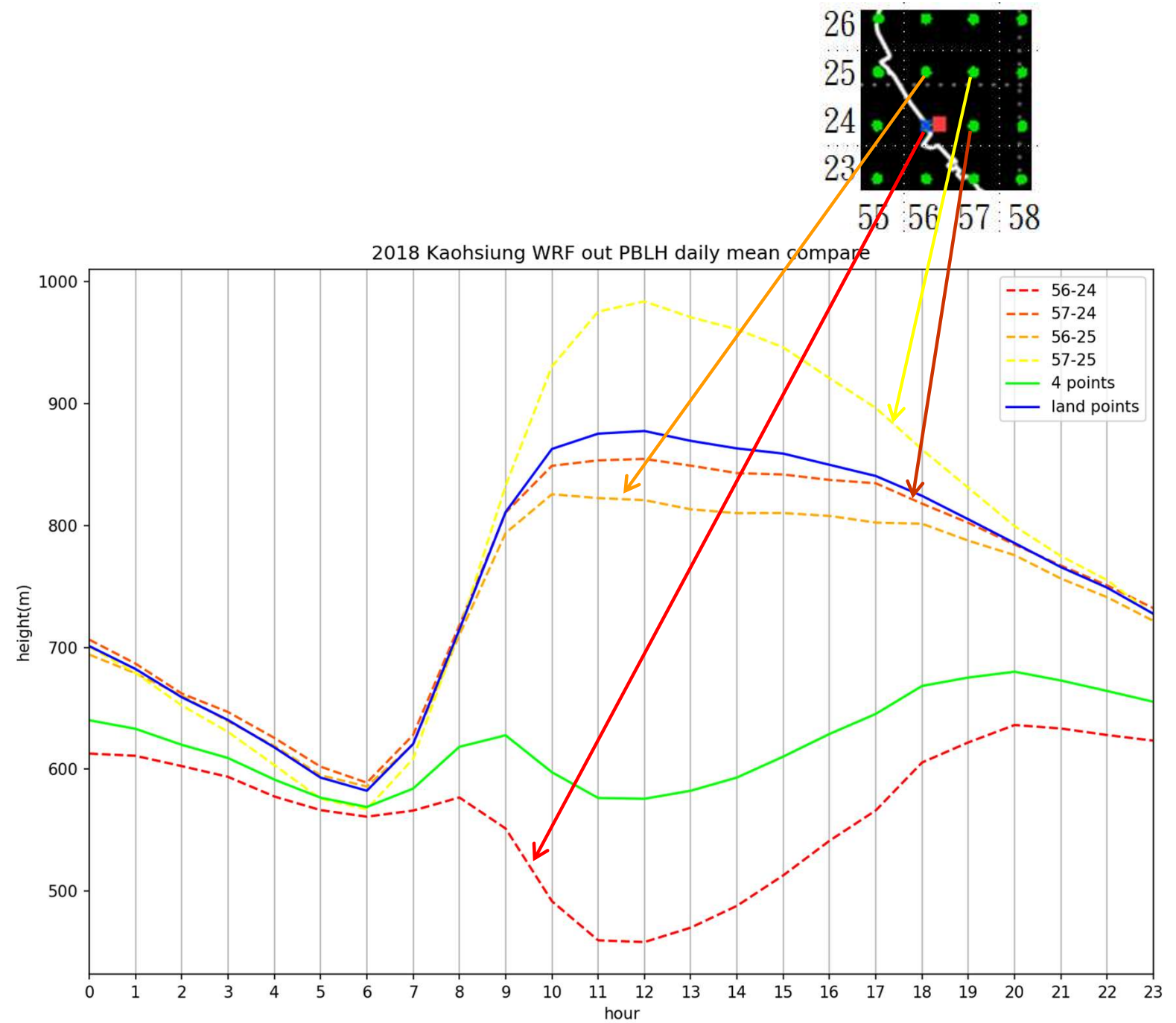
(56, 25)=13 'Urban and Built-Up'

(57, 25)=13 'Urban and Built-Up'



3. PBL height comparison

高雄站： 120.3157E, 22.5659N

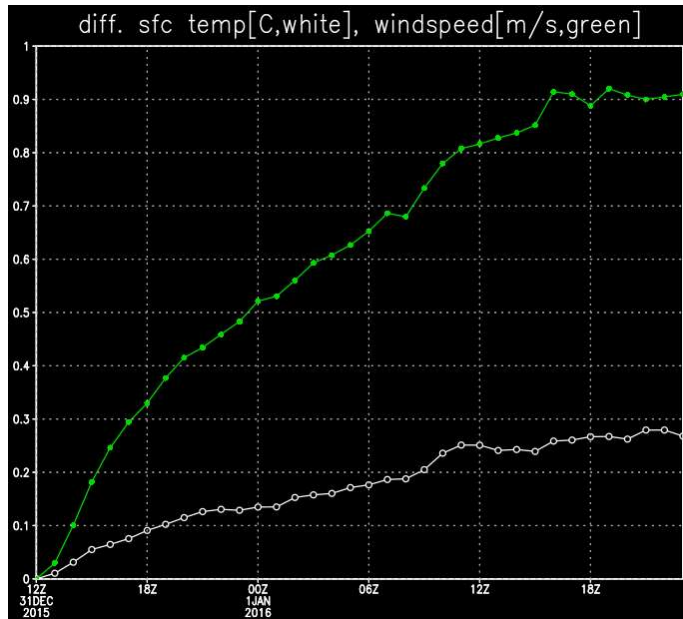


4. Work note

1. [80 CPUs + TS=360] vs. [80 CPUs + TS=240] vs. [200 CPUs + TS=240]
(test 2016 01/01 case, aave(abs(var1-var2)))

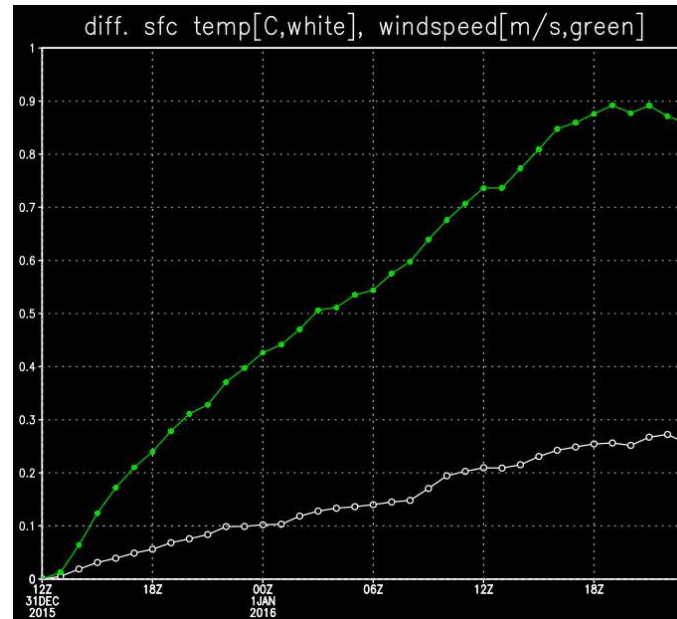
80 CPUs:

TS=360 vs. TS=240

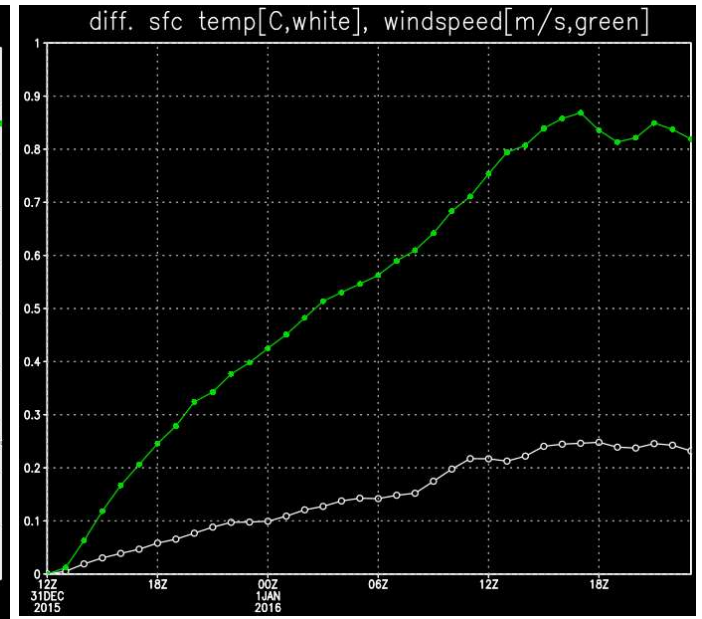


TS=240:

80 CPUs vs. 200 CPUs



80 CPUs vs. 40 CPUs



d01: 91×91, d02: 166×169

d03: 223×223, d04: 223×223

CPUs max.=80 (tile grid more than 10×10)